average, the USO return is 1.4% lower. This is due to the roll, as the cash return during this span was close to zero. The shape of the futures curve—contango or backwardation—determines the sign of the roll return. In addition, large investors forced to roll on pre-determined schedules have to pay large transactions costs. Bessembinder et al. (2012) estimate that ETFs pay about 30 basis points on average per roll, or approximately 4.4% per year in oil futures markets. ETFs need liquidity for the roll, and this is the premium they must pay for it in the futures markets.

5.4. GOLD

Gold is not a good inflation hedge. A case can be made for gold in an investor's diversified portfolio, but it cannot be made on the basis that gold is a real asset.

There are two popular misconceptions about gold: first, that gold is an inflation hedge in terms of how gold moves with inflation (the correlation of gold with inflation); and second, that the long-run performance of gold handily beats inflation (the return of gold, on average, is greater than inflation). Let's examine both. Using data from Global Financial Data, I plot the real price appreciation of gold in Panel A of Figure 11.10 from September 1875 to December 2011. The real price appreciation is defined to be the price of gold divided by CPI. I normalize the index to be 1.0 at the beginning of the sample. The correlation of annual returns of gold and inflation is 23% in Figure 11.10. To repeat: gold is far from a perfect inflation hedge.

If the long-run returns of gold were exactly the same as inflation, then the graph of gold's average real return over time should be exactly a horizontal line. From 1875 to the early 1930s, the price of gold declined in real terms. In 1933 real gold prices suddenly jumped upward when President Franklin Roosevelt signed Executive Order 6102 forbidding the private holdings of gold—one of many attempts to stabilize the financial system during the Great Depression. In 1934 the United States defaulted. The 1934 Gold Reserve Act changed the value of a U.S. dollar from $20.67 per troy ounce to $35. This was an economic default but not legal default. Reinhart and Rogoff (2008) label it an explicit default: the United States reduced the value of its debt payments relative to an external measure of value (at that time, all major currencies were backed by gold).

Real gold prices started rising dramatically in the early 1970s, when the Bretton Woods system of fixed exchange rates broke down. In 1979 and 1980, the real price of gold skyrocketed from below 0.5 to above 3.0 as Fed Chairman Volcker jacked up interest rates to end the Great Inflation. Once inflation stabilized, gold lost much of its value in real terms. Since 2000, however, real gold prices have relentlessly marched upward. At the end of the sample at December 2011, the real price was above 2.0.

The time variation in real gold prices in Panel A of Figure 11.10 unequivocally rejects the hypothesis that gold returns are driven only by inflation. Gold has not